

# Bitrix24 Integration

## 1. Overview

### 1.1 Description

**Bitrix24 Integration** connects **Bitrix24** (CRM and business management platform) with **Newo Platform**.

It enables:

- Checking calendar availability for appointments
- Creating bookings as calendar events with contact linking
- Cancelling appointments by deleting calendar events
- Automatic deal creation at conversation end with employee load balancing
- Creating leads in Bitrix24 CRM
- Creating tasks with LLM-powered payload extraction
- Contact lookup and auto-creation across all flows
- Employee workload balancing for deal assignment

This integration is designed for **service businesses and teams** who use Bitrix24 as their CRM and need **automated scheduling, lead management, and task creation through a conversational AI agent** (voice or chat).

### 1.2 Use Cases

- When a client asks about availability → Query Bitrix24 calendar events and compute free slots
- When a client wants to book → Find or create contact, create calendar event
- When a client wants to cancel → Delete calendar event from Bitrix24
- When a conversation ends → Auto-create deal assigned to the least busy employee
- When a lead arrives (e.g., from Yelp) → Create lead in Bitrix24 CRM
- When a client requests a task → LLM extracts details, creates task in Bitrix24

### 1.3 Supported Features

| Feature                 | Supported | Notes                                              |
|-------------------------|-----------|----------------------------------------------------|
| Check Availability      | ✓         | Calendar-based slot computation with working hours |
| Book Appointment        | ✓         | Calendar event with CRM contact link               |
| Cancel Appointment      | ✓         | Deletes calendar event                             |
| Deal Creation           | ✓         | Auto on session end with employee load balancing   |
| Lead Creation           | ✓         | From Yelp leads or webhook                         |
| Task Creation           | ✓         | LLM-powered payload extraction from conversation   |
| Contact Management      | ✓         | Lookup by phone, auto-create if missing            |
| Employee Load Balancing | ✓         | Assigns deals to employee with fewest open deals   |
| Working Hours Detection | ✓         | Fetches from Bitrix24 calendar settings            |
| Availability Preload    | ✓         | Optional check on conversation start               |

|                        |   |                                       |
|------------------------|---|---------------------------------------|
| Reschedule             | ✗ | Cancel + rebook as workaround         |
| Multi-Calendar Support | ✗ | Single owner calendar per integration |

## 2. Requirements

Before installation:

- Active **Bitrix24** account (cloud or on-premise)
- **Bitrix24 OAuth App** with Client ID and Client Secret
- Calendar configured for the owner user (typically admin, ID=1)

## 3. How to Setup

### 3.1 Step 1 — Authorize in Bitrix24

1. In Newo platform, set `bitrix_base_url`, `bitrix_client_id`, and `bitrix_client_secret`
2. Click the Bitrix24 authorization link provided by the platform
3. Log in to your Bitrix24 account
4. Authorize the Newo app
5. Copy the **authorization code** from the redirect URL
6. Paste the code into the `bitrix_oauth_code` attribute

### 3.2 Step 2 — Connect in Platform

1. Open **Newo** → **Projects**
2. Set the following attributes in **Builder / Attributes**:

| Attribute                                                    | Required | Description                                                                  |
|--------------------------------------------------------------|----------|------------------------------------------------------------------------------|
| <code>bitrix_base_url</code>                                 | ✓        | Bitrix24 instance URL (e.g., <code>https://yourcompany.bitrix24.com</code> ) |
| <code>bitrix_client_id</code>                                | ✓        | OAuth App ID                                                                 |
| <code>bitrix_client_secret</code>                            | ✓        | OAuth App Secret                                                             |
| <code>bitrix_oauth_code</code>                               | ✓        | One-time OAuth authorization code                                            |
| <code>bitrix_owner_id</code>                                 | ✗        | Calendar owner ID (default: 1)                                               |
| <code>bitrix_duration</code>                                 | ✗        | Slot duration in minutes (default: 60)                                       |
| <code>bitrix_availability_days</code>                        | ✗        | Days ahead for availability (default: 1)                                     |
| <code>bitrix_employee_position_name</code>                   | ✗        | Employee position filter for deal assignment (default: cleaner)              |
| <code>bitrix_create_deal_on_conversation_end</code>          | ✗        | Auto-create deal on session end (default: True)                              |
| <code>bitrix_check_availability_on_conversation_start</code> | ✗        | Preload availability on start (default: False)                               |

|                             |   |                                              |
|-----------------------------|---|----------------------------------------------|
| bitrix_enable_slot_check    | ✗ | Enable availability checking (default: True) |
| bitrix_enable_booking       | ✗ | Enable booking (default: True)               |
| bitrix_enable_cancellation  | ✗ | Enable cancellation (default: True)          |
| bitrix_enable_task_creation | ✗ | Enable task creation (default: True)         |
| bitrix_enable_lead_creation | ✗ | Enable lead creation (default: True)         |

3. Click **Save + Publish All**

4. On publish, the SetupFlow automatically:

- Exchanges the OAuth code for access and refresh tokens
- Fetches working hours from Bitrix24 calendar settings
- Creates HTTP connectors for API and token operations
- Registers the task creation tool with the agent
- Injects booking and availability payload schemas

## 4. How to Use

### 4.1 How the Integration Works

The integration uses Bitrix24's **REST API** for calendar events, contacts, deals, leads, tasks, and user management. Availability is computed from calendar events. Deals are auto-created at session end with intelligent employee assignment.

- A **Trigger** is a system event that starts a flow
- An **Action** is the API operation performed in Bitrix24

#### Available Triggers

| Trigger                   | Event                      | Description                                 |
|---------------------------|----------------------------|---------------------------------------------|
| <b>Check Availability</b> | get_available_slots        | When the agent needs to look up open slots  |
| <b>Book Appointment</b>   | book_slot                  | When the agent confirms a booking           |
| <b>Cancel Appointment</b> | convo_agent_cancel_booking | When the agent processes a cancellation     |
| <b>Session Ended</b>      | session_ended              | Auto-create deal on conversation end        |
| <b>Create Task</b>        | create_task                | When agent creates a task from conversation |
| <b>Create Lead</b>        | yelp_new_lead              | When a new lead arrives (e.g., from Yelp)   |
| <b>Conversation Start</b> | conversation_started       | Optional availability preload               |
| <b>Setup</b>              | project_publish_finish     | Initializes integration on deploy           |

#### Available Actions

- **Get Calendar Events** — Fetch events by date range (GET `/rest/calendar.event.get` )
- **Create Calendar Event** — New appointment (POST `/rest/calendar.event.add` )
- **Delete Calendar Event** — Cancel appointment (POST `/rest/calendar.event.delete` )
- **Search Contacts** — Find by phone (POST `/rest/crm.contact.list` )

- **Create Contact** — New CRM contact (POST /rest/crm.contact.add )
- **Create Deal** — New CRM deal with employee assignment (POST /rest/crm.deal.add )
- **Create Lead** — New CRM lead (POST /rest/crm.lead.add )
- **Create Task** — New task with contact link (POST /rest/tasks.task.add )
- **List Users** — Get employees by position (POST /rest/user.get )
- **List Deals** — Get deals for workload counting (POST /rest/crm.deal.list )

## 4.2 Architecture Diagrams

### Overall Sequence

```
sequenceDiagram
    participant U as User
    participant A as ConvoAgent
    participant I as Bitrix24 Integration
    participant T as Bitrix24 OAuth
    participant API as Bitrix24 API

    Note over I,T: Setup (on publish)
    A->>I: project_publish_finish
    I->>T: GET /oauth/token/ (code)
    T-->>I: access_token + refresh_token
    I->>API: GET /rest/calendar.settings.get
    API-->>I: Working hours
    I->>A: Tools registered

    Note over U,API: Availability Check
    U->>A: "Any slots tomorrow?"
    A->>I: get_available_slots (date)
    I->>API: GET /rest/calendar.event.get (owner, date range)
    API-->>I: Calendar events
    I->>I: Compute free slots (subtract booked, slice by duration)
    I->>A: result_success (availability[])

    Note over U,API: Booking
    U->>A: "Book 2:00 PM"
    A->>I: book_slot (customerInfo, dateTime)
    I->>API: POST /rest/crm.contact.list (phone filter)
    alt Contact not found
        I->>API: POST /rest/crm.contact.add (name, phone)
    end
    I->>API: POST /rest/calendar.event.add (type, name, from, to, crm_fields)
    API-->>I: event_id
    I->>A: result_success (booking_id)

    Note over U,API: Session End -> Deal
    A->>I: session_ended
    I->>API: POST /rest/crm.contact.list (phone)
    I->>API: POST /rest/user.get (position filter)
    API-->>I: Employee list
    I->>API: POST /rest/crm.deal.list (stage=NEW)
    API-->>I: Deal counts per employee
```

```

I->>I: Select least busy employee
I->>API: POST /rest/crm.deal.add (title, contact, employee)
API-->>I: deal_id

Note over U,API: Task Creation
U->>A: "Create a task for..."
A->>I: create_task (memory)
I->>I: LLM extract: name, phone, taskTitle
I->>API: POST /rest/crm.contact.list (phone)
I->>API: POST /rest/tasks.task.add (title, deadline, contact)
API-->>I: task_id
I->>A: result_success

```

### AvailabilityFlow

```

flowchart TD
    E["get_available_slots<br>or conversation_started"] --> GATE1["Slot check<br>enabled?"]
    GATE1 -->|No| SKIP1["Skip"]
    GATE1 -->|Yes| API1["GET /rest/calendar.event.get<br>{ownerId, from, to}"]
    API1 --> PARSE1["Extract booked events<br>with start/end times"]
    PARSE1 --> CALC1["Sliding window:<br>generate slots at<br>duration intervals<br>(default: 60 min)<br>skip overlapping"]
    CALC1 --> FORMAT1["Group by date<br>Format as HH:MM AM/PM"]
    FORMAT1 --> OUT1["result_success<br>{availability[]}"]

```

### BookingFlow

```

flowchart TD
    E["book_slot"] --> GATE1["Booking<br>enabled?"]
    GATE1 -->|No| SKIP1["Skip"]
    GATE1 -->|Yes| CONTACT1["POST /rest/crm.contact.list<br>{phone filter}"]
    CONTACT1 --> FOUND1["Contact<br>found?"]
    FOUND1 -->|No| CREATE1["POST /rest/crm.contact.add<br>{name, phone}"]
    CREATE1 --> CID1["contact_id"]
    FOUND1 -->|Yes| CID1
    CID1 --> EVENT1["POST /rest/calendar.event.add<br>{type: user,<br>name: Client Appointment,<br>from/to ISO times,<br>crm_fields: [C_contact_id]}"]
    EVENT1 --> OUT1["result_success<br>{booking_id}"]

```

### DealFinalizationFlow (on session end)

```

flowchart TD
    E["session_ended"] --> GATE1["Deal creation<br>enabled?"]
    GATE1 -->|No| SKIP1["Skip"]
    GATE1 -->|Yes| BOOK1["Has bookings<br>in persona?"]
    BOOK1 -->|No| SKIP2["Skip"]
    BOOK1 -->|Yes| CONTACT1["POST /rest/crm.contact.list<br>{phone}"]
    CONTACT1 --> EMP1["POST /rest/user.get<br>{ACTIVE: Y,<br>position filter}"]

```

```

EMP --> DEALS["POST /rest/crm.deal.list<br>{stage: NEW}"]
DEALS --> BEST["Count deals per employee<br>Select least busy"]
BEST --> DEAL["POST /rest/crm.deal.add<br>{title: service + date,<br>contact,
employee,<br>stage: NEW,<br>source: CHAT}"]
DEAL --> OUT["Deal created"]

```

## TaskFlow

```

flowchart TD
    E["create_task"] --> LLM["LLM extract from memory:<br>firstName, lastName,
<br>phoneNumber, taskTitle<br>(gemini25_flash)"]
    LLM --> CONTACT["POST /rest/crm.contact.list<br>{phone}"]
    CONTACT --> FOUND["Contact?"]
    FOUND -->|"No"| SKIP_LINK["No CRM link"]
    FOUND -->|"Yes"| LINK["UF_CRM_TASK:<br>[C_contact_id]"]
    SKIP_LINK --> TASK["POST /rest/tasks.task.add<br>{title, deadline,
<br>created_by: 1,<br>responsible_id: 1}"]
    LINK --> TASK
    TASK --> OUT["result_success"]

```

## CancellationFlow

```

flowchart TD
    E["convo_agent_cancel_booking"] --> GATE["Cancellation<br>enabled?"]
    GATE -->|"No"| SKIP["Skip"]
    GATE -->|"Yes"| DEL["POST /rest/calendar.event.delete<br>{id: booking_id}"]
    DEL --> OUT["result_success"]

```

## 4.3 Where to Test

Testing can be done through the **Newo conversation interface** (chat or voice). Major flows (availability, booking, cancellation, lead, task) have dedicated webhook test endpoints.

## 4.4 Testing and Verification

### To test the integration:

1. **Setup:** Publish and verify OAuth token exchange + working hours fetch
2. **Availability:** Ask about open slots — verify calendar events are queried and free times computed
3. **Booking:** Complete a booking — verify calendar event created in Bitrix24 with contact link
4. **Cancellation:** Cancel a booking — verify calendar event deleted
5. **Deal:** End a conversation after booking — verify deal created and assigned to least busy employee
6. **Task:** Ask agent to create a task — verify task appears in Bitrix24 with contact link

### If no action occurs:

- Ensure OAuth credentials are set ( `bitrix_base_url` , `bitrix_client_id` , `bitrix_client_secret` )
- Verify `bitrix_access_token` was obtained
- Confirm `bitrix_owner_id` points to a valid user with a calendar
- Check feature flags are enabled for the desired operations

- Verify `bitrix_employee_position_name` matches actual employee positions (for deal assignment)
- Re-publish the project if credentials were changed

*Note: This integration creates real records in Bitrix24. Calendar events, deals, leads, tasks, and contacts are reflected in the live CRM.*

## 5. FAQ

**Q: How does employee load balancing work for deals?** A: When a deal is created at session end, the integration fetches all active employees matching `bitrix_employee_position_name`, counts their open deals (stage=NEW), and assigns the new deal to the employee with the fewest deals.

**Q: What is the difference between deals, leads, and tasks?** A: **Deals** are auto-created when a conversation ends (if a booking was made) — they represent a business opportunity with employee assignment. **Leads** are created from external sources (e.g., Yelp) as initial contact records. **Tasks** are created on demand from conversation context with LLM-extracted details.

**Q: How is availability computed?** A: The integration queries the calendar owner's events for the date range, then generates time slots at `bitrix_duration` intervals (default: 60 min), removing slots that overlap with existing events.

**Q: Can I connect to an on-premise Bitrix24?** A: Yes. Set `bitrix_base_url` to your on-premise instance URL. The OAuth token endpoint remains at `https://oauth.bitrix.info/oauth/token/` for all Bitrix24 installations.

**Q: How does the token refresh work?** A: On HTTP 401 responses, the integration automatically refreshes the token via `https://oauth.bitrix.info/oauth/token/` with the stored refresh token. A separate HTTP connector ( `bitrix_token_connector` ) handles token operations.

**Q: What information does the LLM extract for tasks?** A: Gemini 2.5 Flash extracts `firstName`, `lastName`, `phoneNumber` (E.164 format), and `taskTitle` (max 10 words) from the conversation memory. If last name is not available, the first name is copied.

**Q: Can I disable deal creation at session end?** A: Yes. Set `bitrix_create_deal_on_conversation_end` to `False`. This disables the DealFinalizationFlow entirely.